

# JIAHAO TIAN

Hangzhou, Zhejiang, China

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🌐 [jiahaotian-stat.github.io](https://jiahaotian-stat.github.io)

🐙 [github.com/jiahaotian-stat](https://github.com/jiahaotian-stat)

## Education

School of Mathematical Sciences, Zhejiang University

Aug. 2023 – Jun. 2027 (Expected)

*Bachelor of Science in Statistics*

*Hangzhou, Zhejiang, China*

- **GPA:** 4.01/4.30
- **Current Coursework:** Advanced Probability Theory (Grad), Advanced Mathematical Statistics (Grad), and Matrix Computations (Grad).

## Selected Coursework

Probability Theory (92)

Mathematical Statistics (95)

Stochastic Processes (99)

Real Analysis (87)

Functional Analysis (92)

Generalized Functions (95)

Regression Analysis (96)

Asymptotic Statistics (PhD, 95)

Stochastic Differential Equations (Grad, 96)

## Research Experience

Heavy-Tailed Robustness of Multiple-Knockoff Aggregation Methods

Sep. 2025 – Present

*Undergraduate Researcher | Advised by Prof. Lijun Wang*

*Hangzhou, China*

- Studied the robustness of multiple knockoff aggregation methods under Gaussian and heavy-tailed misspecification, identifying substantial dataset heterogeneity under Cauchy designs, including cases of aggregation power loss despite effective single-run knockoffs.
- Developed an asymptotic explanation using **regular variation**, null sign symmetry, concentration arguments, and **Extreme Value Theory** to study threshold inflation and declining signal-selection probabilities in sparse high-dimensional regimes.
- Explored robust feature statistics and developed a compute-once, evaluate-many procedure for efficiently tuning the internal knockoff threshold across multiple aggregation settings.

Distributional Balancing Weights for Conformal Calibration under Covariate Shift

Mar. 2026 – Present

*Research Assistant | Advised by Prof. Guanhua Chen (UW-Madison)*

*Remote*

- Developed a **DB-CC** framework for covariate shift, using **balance-transfer pairs** to connect covariate distributional balance with conformity-score CDF estimation, and specialized it to **characteristic function distance** balancing.
- Contributed to proofs of **finite-sample score-distribution and coverage bounds**, using asymptotic and harmonic analysis to study how kernel spectra and score-function complexity affect calibration performance.
- Designed experiments to validate the theory, studying how score-function complexity, kernel geometry, and regularization tuning affect calibration performance, and applied DB-CC to **Split CP**, **CQR**, and **UACQR**.
- **Guanhua Chen\*** & **Jiahao Tian\*** (co-first, co-corresponding; alphabetical order), *under review at ICLR 2027*.

## Academic Writing & Presentations

Academic Notes Repository | [jiahaotian-stat.github.io/Notes](https://jiahaotian-stat.github.io/Notes)

Spring 2026 – Present

- Maintained a repository of course notes and seminar materials in mathematics and statistics.

Convergence of Random Series and Large Deviations

Fall 2025

*Advanced Probability Theory Seminar, Zhejiang University*

*Hangzhou, China*

Robustness of Multiple Knockoff Methods

Fall 2025

*Statistics Research Seminar, Zhejiang University*

*Hangzhou, China*

Distributional Balancing Weights for Conformal Calibration under Covariate Shift

Fall 2026

*Statistics Research Seminar, Zhejiang University*

*Hangzhou, China*

## Teaching Experience

Teaching Assistant, Probability and Mathematical Statistics

Sep. 2026 – Present

*School of Mathematical Sciences, Zhejiang University*

*Hangzhou, China*

## Skills & Tests

**Programming:** Python, R, Shell, C/C++, Julia, MATLAB

**Tools:** Linux, Git/GitHub, L<sup>A</sup>T<sub>E</sub>X, Quarto, Typst, MkDocs

**Standardized Tests:** GRE Quantitative: 170/170 | TOEFL iBT: 5.5/6.0